SiteSaathi — Mobile App Updates & Architecture

Official Engineering & Feature Release Documentation • Offline-First Android APK

Version: 1.0.0 (Capacitor Android)	**Application Type:** 100% Offline Mobile App
Build Date: 15 October 2026	**Output Target:** standalone APK (4.68 MB)
Technology Stack: React 19, Vite 8, Dexie, Capacitor 8	**Primary Theme:** Unified Cobalt Blue (#2563EB)

1. Executive Overview & Transformation Goal

SiteSaathi has been completely re-architected from a promotional web showcase into a dedicated, **100% offline-first Android mobile application**. Construction sites and remote field locations frequently suffer from weak or nonexistent mobile network connectivity. With this update, **all data is stored and indexed locally on the user's mobile device** using Dexie (IndexedDB on Android WebView). The user incurs zero cloud costs, zero database hosting fees, and has guaranteed instant data access at all times.

2. Summary of Updates Implemented

Module	Previous Web State	New Mobile App Implementation
App Shell	Opened wide marketing landing page with 'Showcase' banners and 'Live Preview' links.	Opens directly into the Mobile App. Marketing landing page permanently removed.
Desktop / Laptop View	Stretched across wide 1920px screen like a desktop web SaaS dashboard.	Renders a realistic Smartphone Device Simulator (Status Bar, Dynamic Island, 390px frame).
Authentication & Trade Locking	Generic multi-trade switcher allowed live switching between trades on the header.	Fixed trade binding upon registration/login (Civil, Plumbing, Electrical). Switcher removed.
Top App Header	8 horizontal buttons crowded into header, overflowing and clipping off-screen.	Streamlined 52px Mobile Top Bar with user avatar, trade badge, language, bell, and logout.
Dashboard KPI Cards	4 columns squished into 80px cards with truncated labels ('Worke...', 'Overd...').	Mobile-first 2x2 Grid (2 columns, 2 rows) with large readable stats and zero text truncation.
Navigation Model	Horizontal desktop navigation tabs with hidden mobile elements on large screens.	Native Mobile Bottom Navigation Bar (5 tabs) + Floating Action Button (+) always docked.
Android Packaging	Browser web page only.	Capacitor 8 Android native project with one-command debug APK compilation.

3. Technical Architecture & Offline Storage

1. Local Persistence (IndexedDB on Android WebView):
All data models—including Sites, Tools, Material Deliveries, Machinery Logs, Labour Registers, and Pending Dues—are managed through Dexie.js (IndexedDB). When compiled into an Android APK, Android's WebView persists IndexedDB directly into the device's private app sandbox (/data/user/0/com.sitesaathi.app/app_webview/IndexedDB). No network requests are ever required for CRUD operations.

2. Capacitor Native Bridge:
Capacitor packages the React/Vite production distribution bundle directly inside Android assets (android/app/src/main/assets/public). When the APK is opened, assets load locally with 0ms server latency. Mixed content and cleartext flags are explicitly configured in capacitor.config.ts.

4. Operational Workflow for Field & Developer Testing

A. Development on Laptop (Instant Live Updates):
Run the development server on your laptop with hot module reloading. Open your browser to test updates instantly in the phone simulator:

```
npm run dev # Live at http://localhost:5173 with Phone Simulator
```

B. Compiling the Updated Android APK:
Whenever updates are completed on the laptop and you wish to generate a fresh installable APK for your Android mobile phone, run:

```
npm run apk:build
```

This automated command builds the React bundle, copies assets into the Android native directory, and executes Gradle to output the signed debug APK file.

C. Installing on a Connected Android Device:
To push the freshly built APK directly to a USB-connected phone with developer mode enabled:

```
& "C:\Users\vishu\AppData\Local\Android\Sdk\platform-tools\adb.exe" install -r
"android\app\build\outputs\apk\debug\app-debug.apk"
```

5. Deliverables & File Locations

- **Android APK:** android/app/build/outputs/apk/debug/app-debug.apk (4.68 MB)
- **Capacitor Config:** capacitor.config.ts (App ID: com.sitesaathi.app)
- **Android Native Project:** android/ (Gradle 8.14, SDK 36, Java 21)
- **Mobile Device Shell:** src/components/common/MobileDeviceShell.tsx
- **Mobile Header Bar:** src/components/common/Header.tsx
- **2x2 Mobile Dashboard:** src/components/dashboard/DashboardOverview.tsx
- **Touch Bottom Navigation:** src/components/common/BottomNav.tsx

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